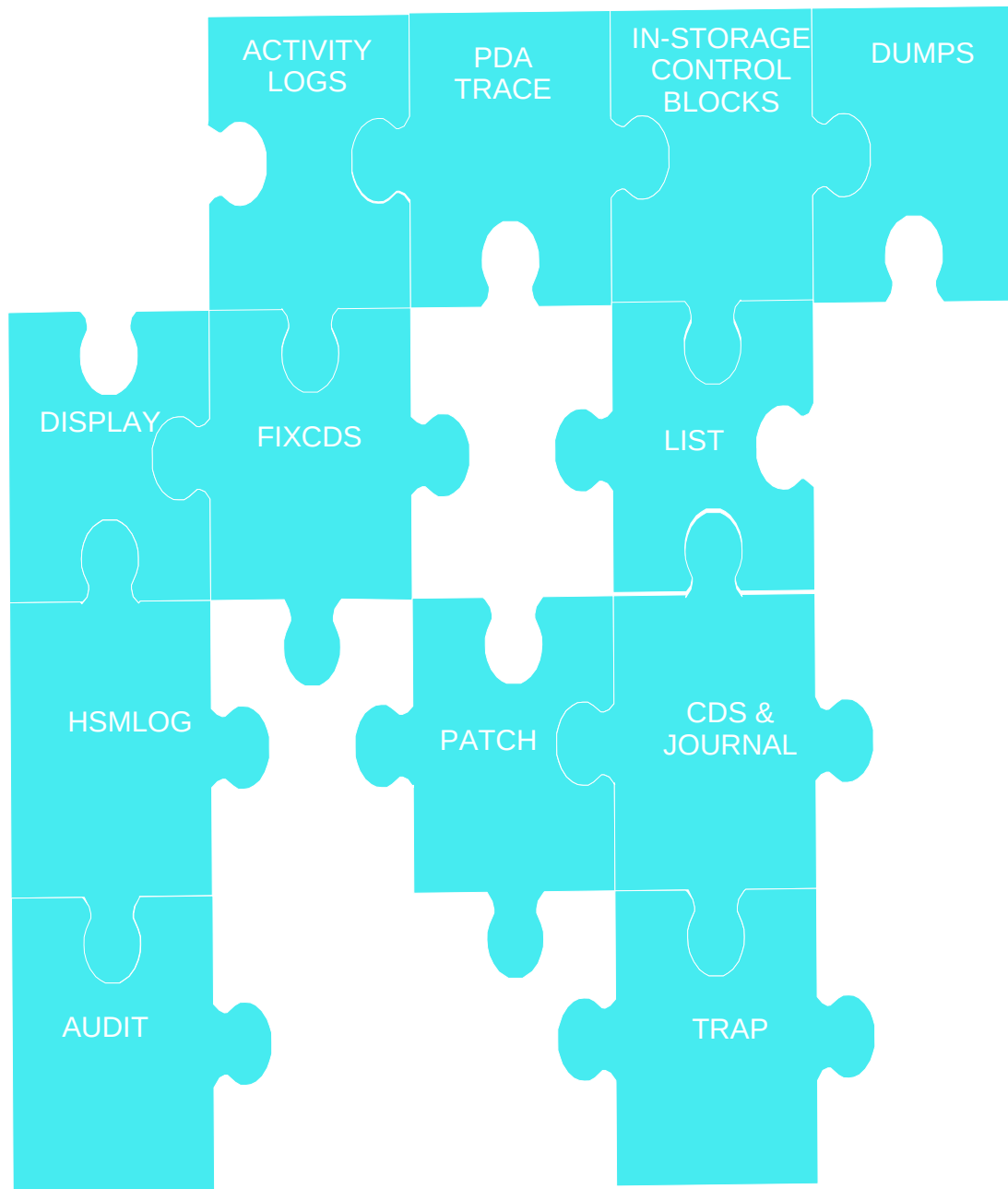


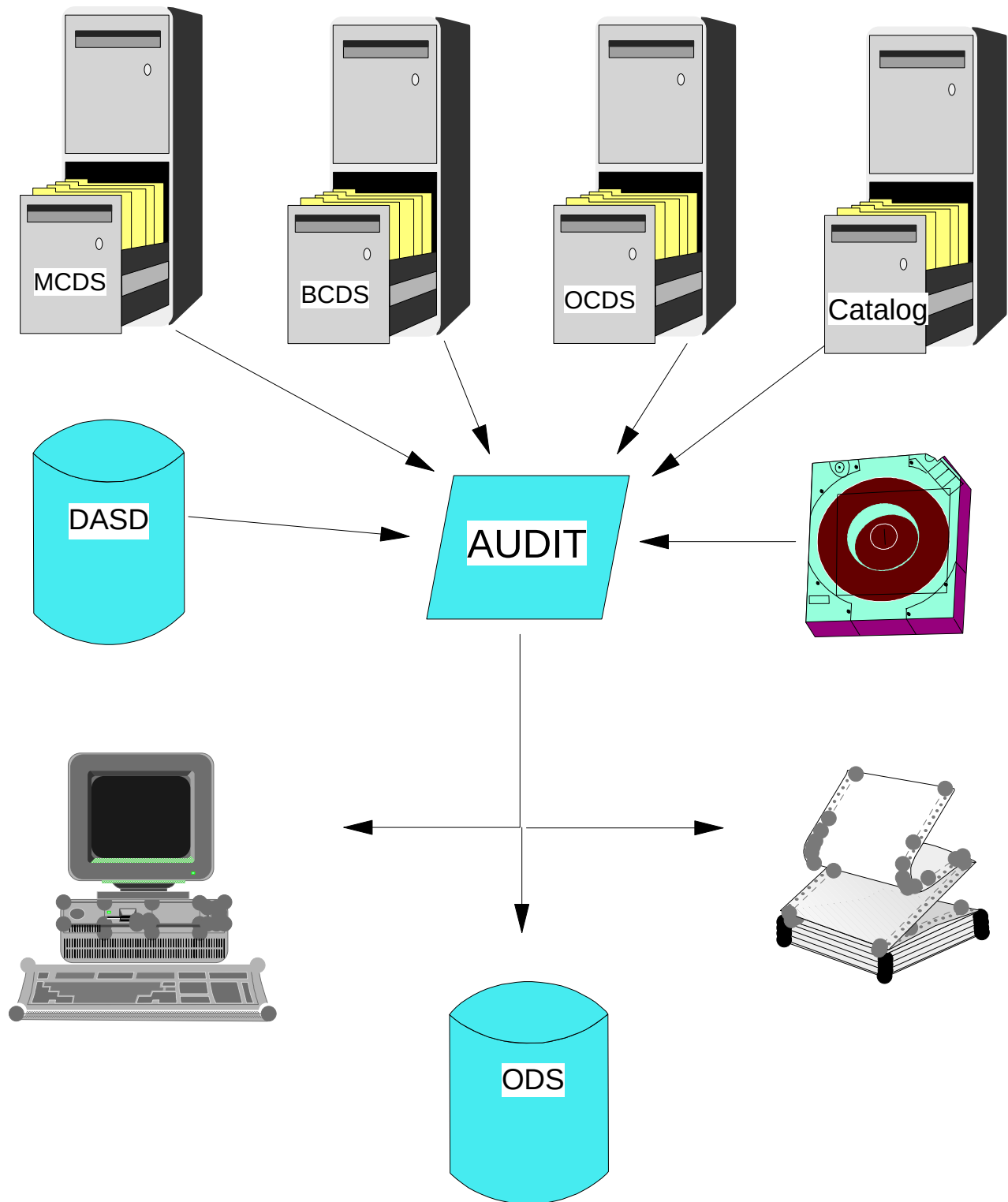




Objectives

- Understand the AUDIT command's functions and capabilities as a diagnostic and corrective tool
- Be able to use the AUDIT command to detect and correct errors in the control data sets

The AUDIT Command



The AUDIT command is used to detect, diagnose, and provide repairs for discrepancies between VTOCs, catalogs, and the control data sets. The following are characteristics of the AUDIT command:

- creates and optionally executes FIXCDS commands
- can only be executed by an HSM authorized user
- a long running command, i.e. only one long running command can be executing at given time. The long running commands are: AUDIT, EXPIREBV, RECYCLE, REPORT, and TAPECOPY.
- serial execution

Notes:

New vs. Old

New

Old

ABARSCONTROLS

DATASETNAMES

VOLUMECONTROLS(MIGRATION)

VOLUMES

VOLUMECONTROLS(BACKUP)

BACKUPVOLUMES
BACKUPTYPE

MASTERCATALOG
USERCATALOG

DATASETCONTROLS(MIGRATION)

MIGRATIONCONTROLDATASET

DATASETCONTROLS(BACKUP)

BACKUPCONTROLDATASET

DIRECTORYCONTROLS

OFFLINECONTROLDATASET

MEDIACONTROLS(SDSP)

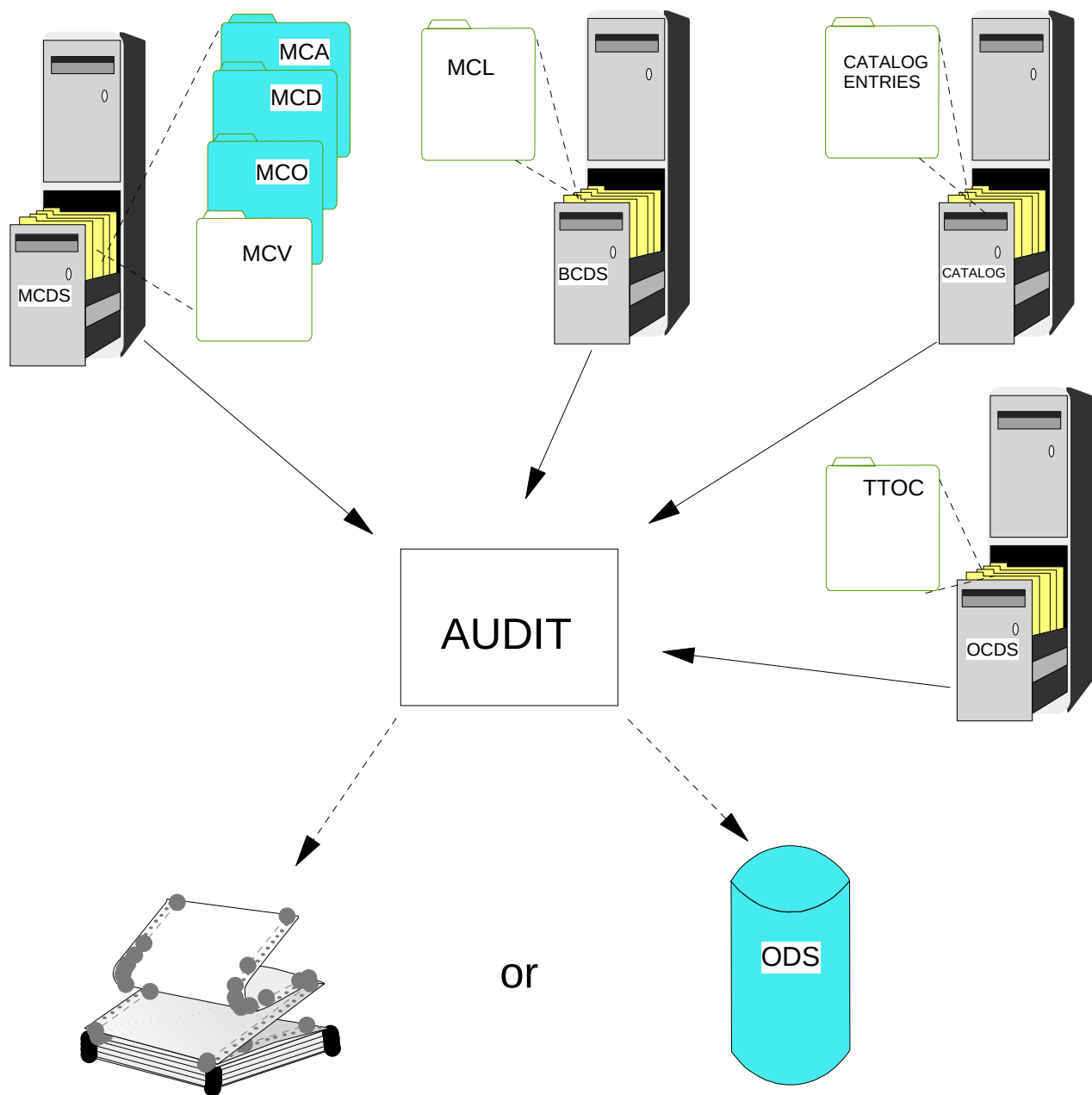
ALL

MEDIACONTROLS VOLUMES

Except for AUDIT MASTERCATALOG and AUDIT USERCATALOG, it is highly recommended that only the new commands be used. Significant changes have been made regarding the new commands that improve their effectiveness and efficiency.

Notes:

AUDIT DATASET CONTROLS (MIGRATION)



AUDIT DATASETCONTROLS(MIGRATION) DATASETNAMES(dsname) |
LEVELS(qualifier) | RESUME
FIX | NOFIX
OUTDATASET(dsname) | SYSOUT(class)
REPORT(ALL | ERRORS)
SERIALIZATION(DYNAMIC | CONTINUOUS)

The MCDS is read sequentially and the following is performed:

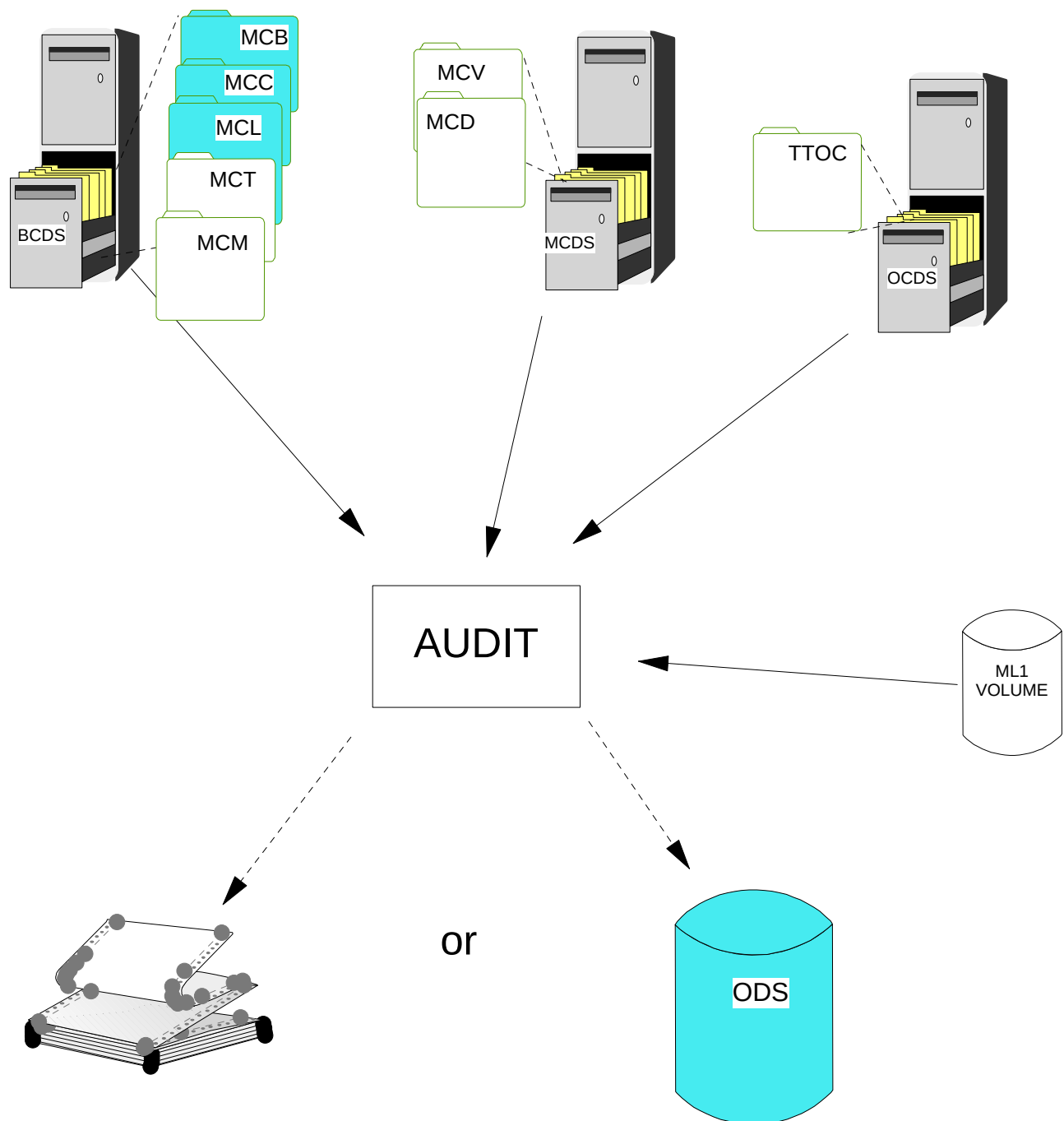
- The following checks are made for each MCD record that is found:
 - Dataset properly cataloged to MIGRAT
 - MCA record (if not in the SDSP)
 - MCO record (if VSAM)
 - MCL record (if in need of backup)
 - TTOC record (if migrated to ML2 tape)
 - MCV record
- The following check is made for each MCA record that is found:
 - MCD record
- The following check is made for each MCO record that is found:
 - MCA record(s)

When to use this command:

- after a recovery of the BCDS or OCDS
- after the recovery of a catalog
- after recall failures
- periodically even when no errors are detected

Notes:

AUDIT DATASET CONTROLS (BACKUP)



AUDIT DATASETCONTROLS(BACKUP) DATASETNAMES(dsname) |
LEVELS(qualifier) | RESUME
FIX | NOFIX
OUTDATASET(dsname) | SYSOUT(class)
REPORT(ALL | ERRORS)
SERIALIZATION(DYNAMIC | CONTINUOUS)

The BCDS is read sequentially and the following is performed:

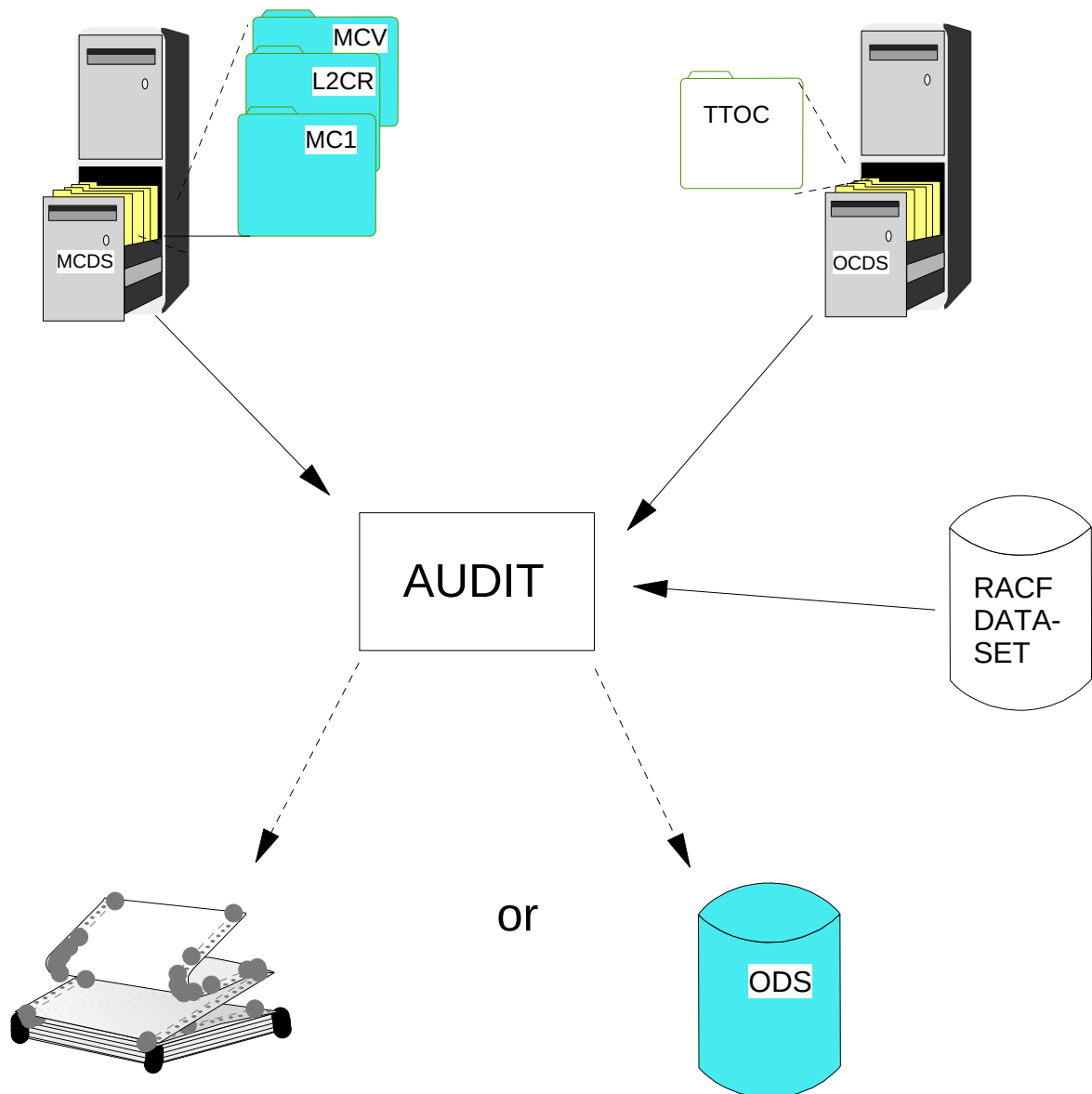
- The following checks are made for each MCB record that is found:
 - All corresponding MCC records
 - MCT record
 - MCV record (if on ML1) and ML1 VTOC entry
 - MCM record
 - TTOC record
- The following check is made for each MCL record that is found:
 - MCD record
- The following check is made for each MCC record that is found:
 - MCB record

When to use this command:

- after a recovery of the MCDS or OCDS
- after the recovery of an ML1 volume
- after recover failures
- periodically even when no errors are detected

Notes:

AUDIT VOLUMECONTROLS (MIGRATION)



AUDIT VOLUMECONTROLS(MIGRATION)
VOLUMES(volser)
FIX | NOFIX
OUTDATASET(dsname) | SYSOUT(class)
REPORT(ALL | ERRORS)
SERIALIZATION(DYNAMIC | CONTINUOUS)

The MCV records are read sequentially and the following is performed:

- TTOC record (if a tape volume)
 - Verify RACF protect flags are consistent between MCV and TTOC records.
- RACF TAPEVOL class

The L2CR records are read sequentially and the following is performed:

- MCV record
- Verify that MCV record indicates DASD device type

The MC1 records are read sequentially and the following is performed:

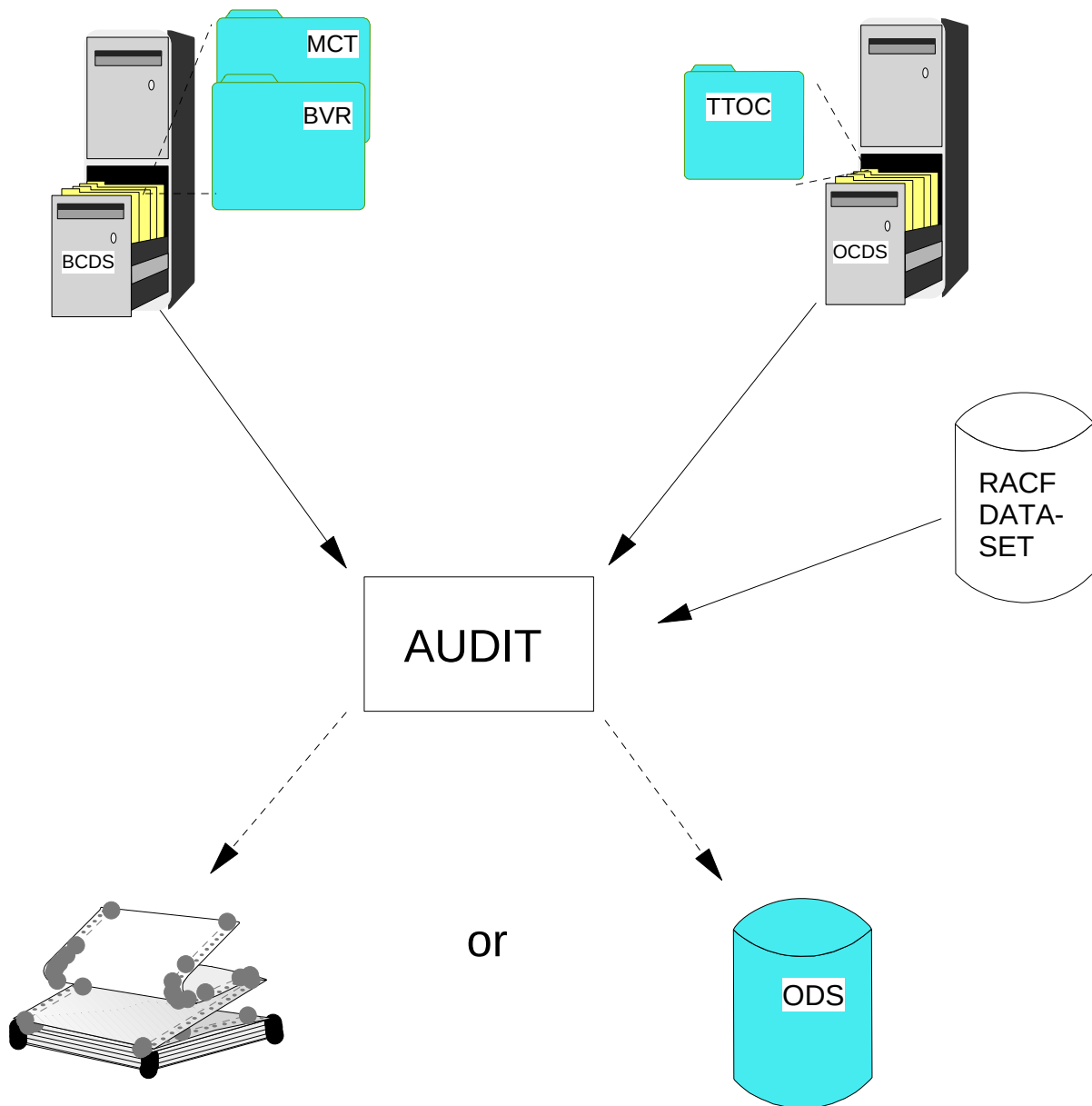
- Verify MC1 record extensions are present
- Verify that there are no extraneous MC1 extensions

When to use this command:

- after a recovery of the OCDS
- after the recovery of the RACF database

Notes:

AUDIT VOLUME CONTROLS (BACKUP)



```
AUDIT VOLUMECONTROLS(BACKUP)
      VOLUMES( volser...) | BACKUPTYPE(DAILY(day) | SPILL | ALL
      FIX | NOFIX
      OUTDATASET(dsname) | SYSOUT(class)
      REPORT(ALL | ERRORS)
      SERIALIZATION(DYNAMIC | CONTINUOUS)
```

The MCT records are read sequentially and the following checks are performed:

- TTOC records
- RACF data set for the TAPEVOL class
- BVR consistencies

The BVR records are read sequentially and the following checks are performed:

- MCT records
- BVR checked for correct number of extensions

In addition, the TTOC records in the OCDS related to backup tapes are read sequentially and the following checks are made:

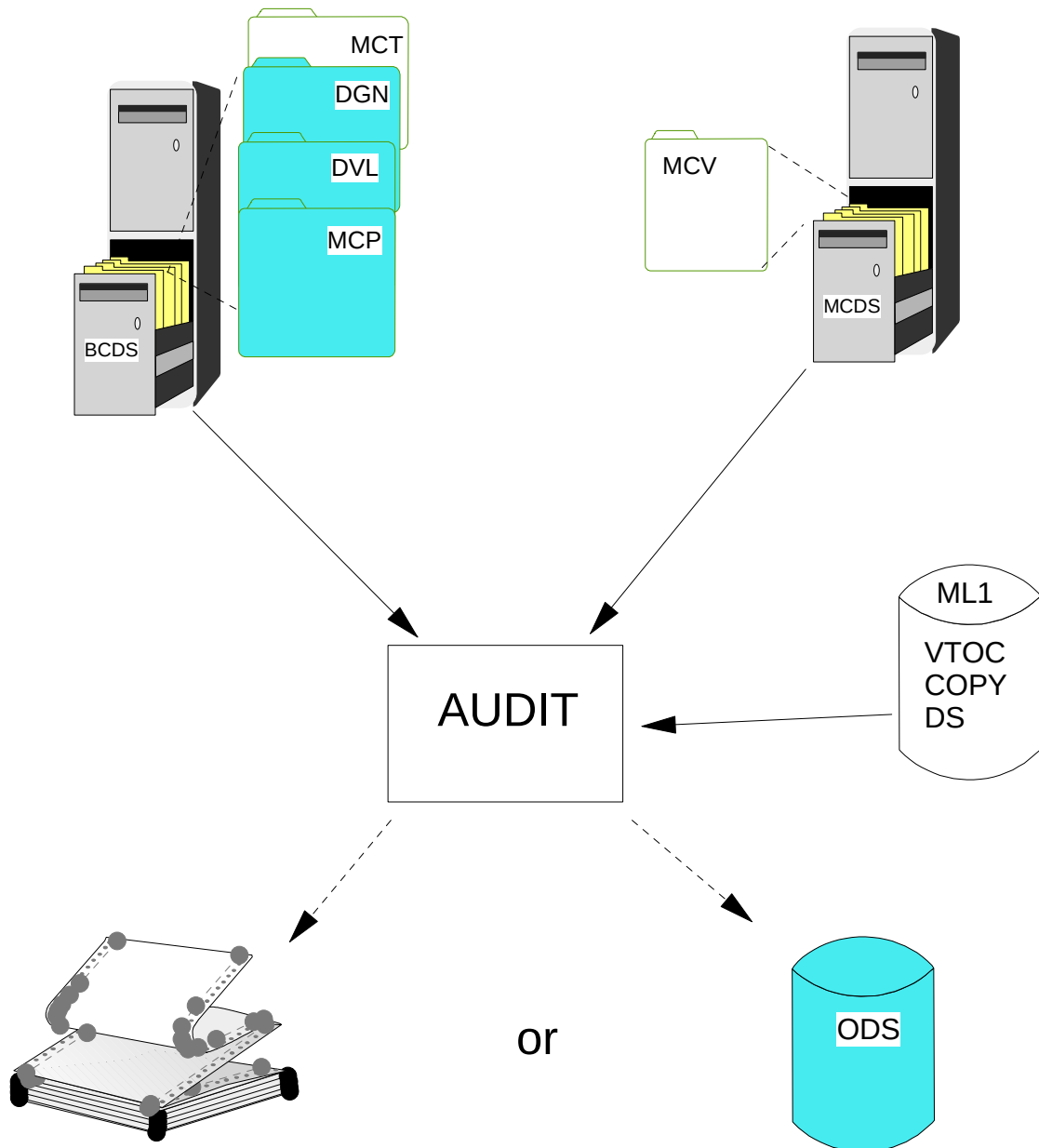
- MCT records

When to use this command:

- After recovery of the OCDS
- After recovery of the RACF database

Notes:

AUDIT VOLUME CONTROLS (RECOVERABLE)




```
AUDIT VOLUMECONTROLS(RECOVERABLE) VOLUMES( volser...) |  
      FIX | NOFIX  
      OUTDATASET(dsname) | SYSOUT(class)  
      REPORT(ALL | ERRORS)  
      SERIALIZATION(DYNAMIC | CONTINUOUS)
```

The MCP record is read to ensure that it exists and the following check is performed:

- MCP record
 - Must exist
- If the MCP record indicates the volume has been backed up, the following checks are made for the VTOC copy dataset.
 - If stored on a migration volume, ensure that the MCP record exists.
 - If stored on a DASD backup volume, ensure that the MCT record exists.
 - If stored on a tape backup volume, ensure that the MCT records for each tape exist.
- If the MCP record indicates the volume has been dumped, the associated DGN records are read and the following checks made:
 - DGN exists
 - DVL records pointed to by DGN exist and point back to DGN, contains the same dumpclass as the DGN record, and has the same sequence number as indicated in the DGN record.

If the command is not limited to a specific volume, the DGN record is read sequentially and the following checks are performed:

- MCP record
 - Must exist and must point back to DGN record

If the command is not limited to a specific volume, the DVL record is read sequentially and the following checks are performed:

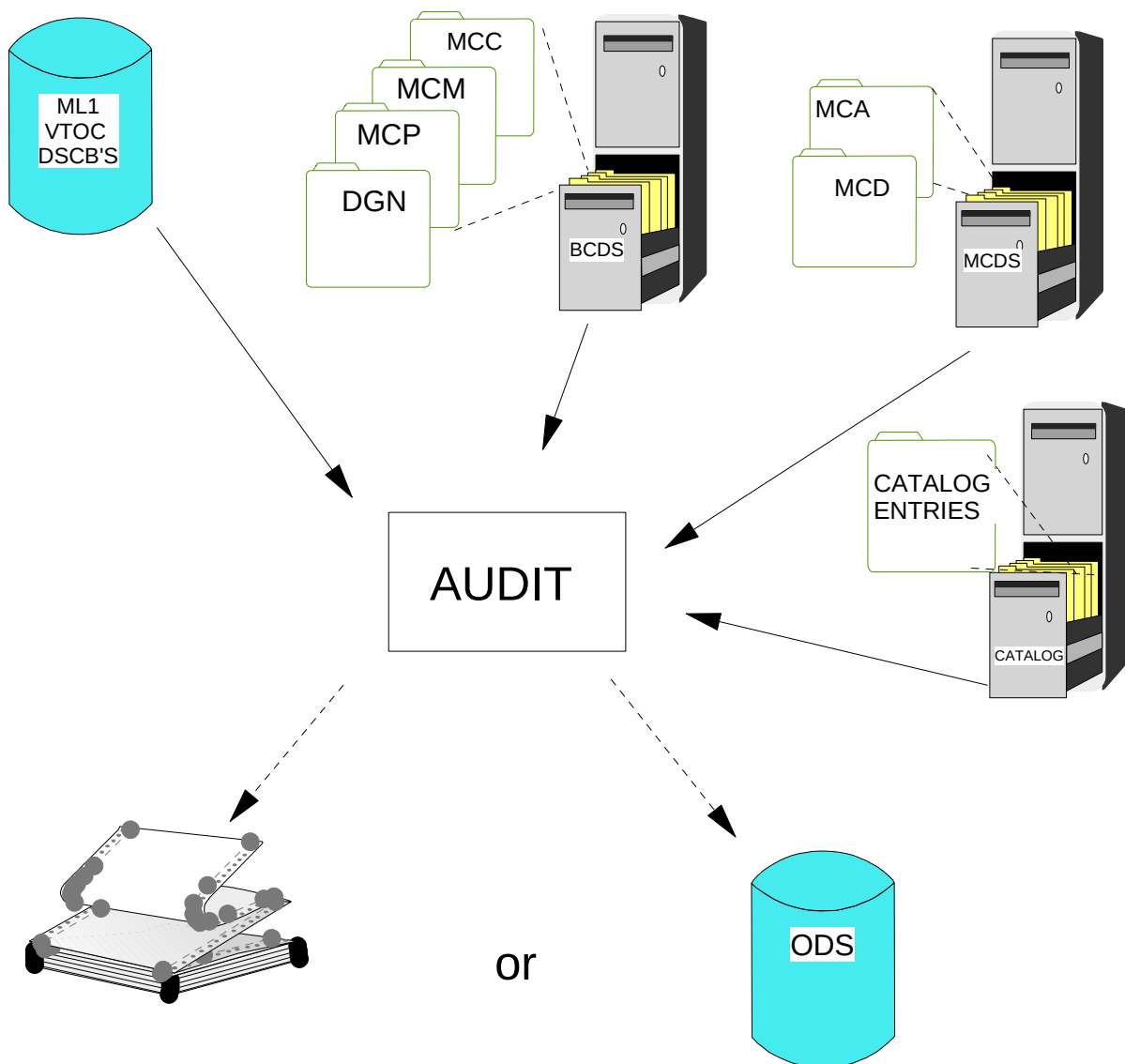
- DGN record
 - Must contain the same dumpclass
 - Dump volume must be part of the generation
 - Volume sequence number must match that recorded in DGN

When to use this command:

- After recovery of the MCDS
- After recovery of the BCDS

Notes:

AUDIT DIRECTORYCONTROLS VOLUMES (ml1vols...)



```
AUDIT DIRECTORYCONTROLS VOLUMES(ml1vols ... ) |  
      FIX | NOFIX  
      OUTDATASET(dsname) | SYSOUT(class)  
      REPORT(ERRORS)  
      SERIALIZATION(DYNAMIC | CONTINUOUS)
```

The ML1 VTOC is read and the following is performed:

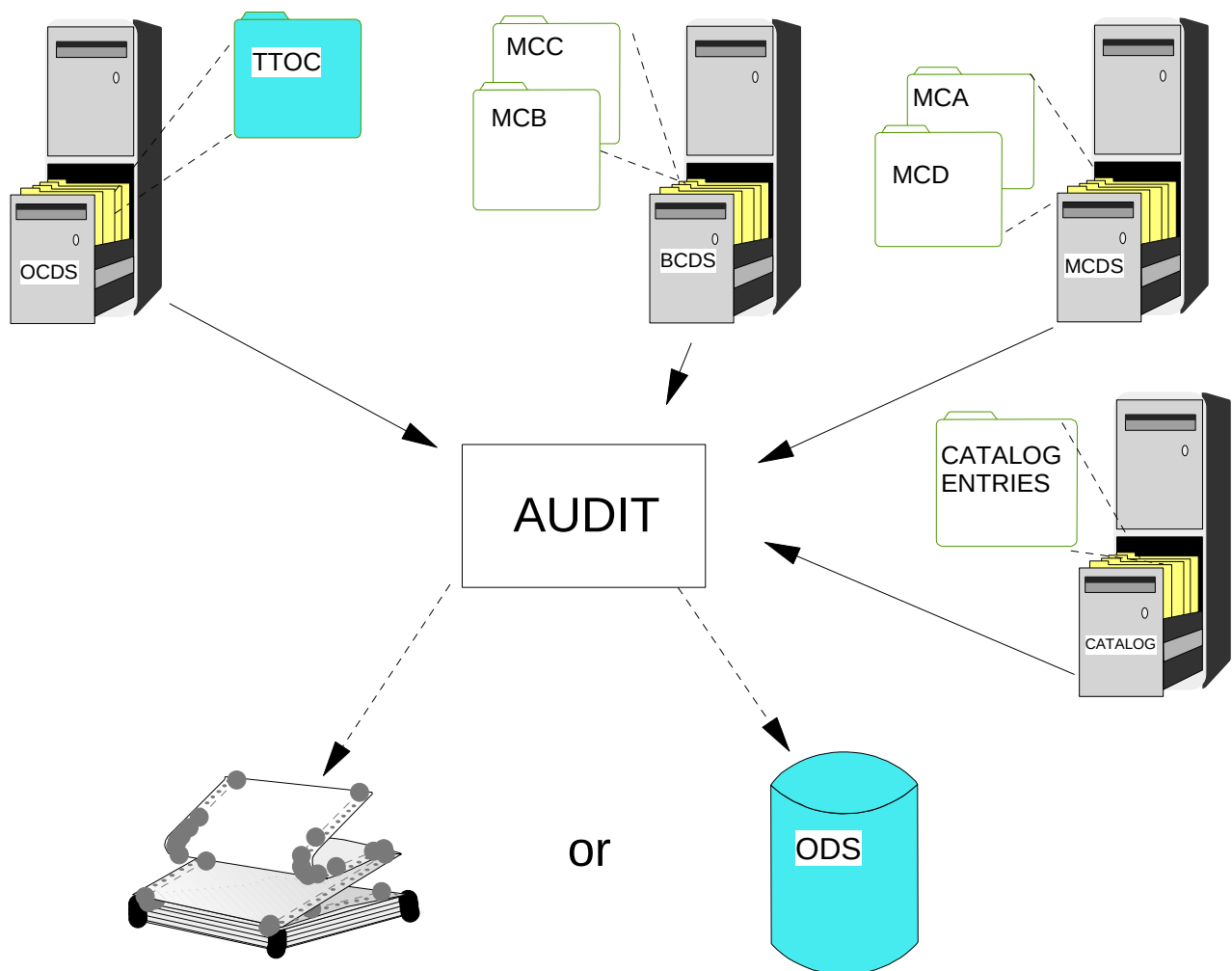
- If a backup version is found:
 - MCM record
 - MCC record
- If a VTOCCOPY or a VCATCOPY is found:
 - MCP record, exists and indicates that the volume has been backed up.
- If a DUMP VTOCCOPY is found:
 - MCP record, exists and indicates that the volume has been dumped.
 - At least one DGN record as indicated by the MCP record must point to the dump VTOC copy dataset.
- If a migrated version is found:
 - MCA record exists
 - MCD record exists
 - Catalog entry indicates MIGRAT
- If an empty HSM-owned data set is found, scratch it

When to use this command:

- After recovery of the BCDS or MCDS
- After recovery of an ML1 volume

Notes:

AUDIT DIRECTORYCONTROLS VOLUMES(tapevol1...)



```
AUDIT DIRECTORYCONTROLS VOLUMES(tapevol1 ... ) |  
      FIX | NOFIX  
      OUTDATASET(dsname) | SYSOUT(class)  
      REPORT( ERRORS)  
      SERIALIZATION(DYNAMIC | CONTINUOUS)
```

The following checks are performed for each TTOC record related to the specified volume:

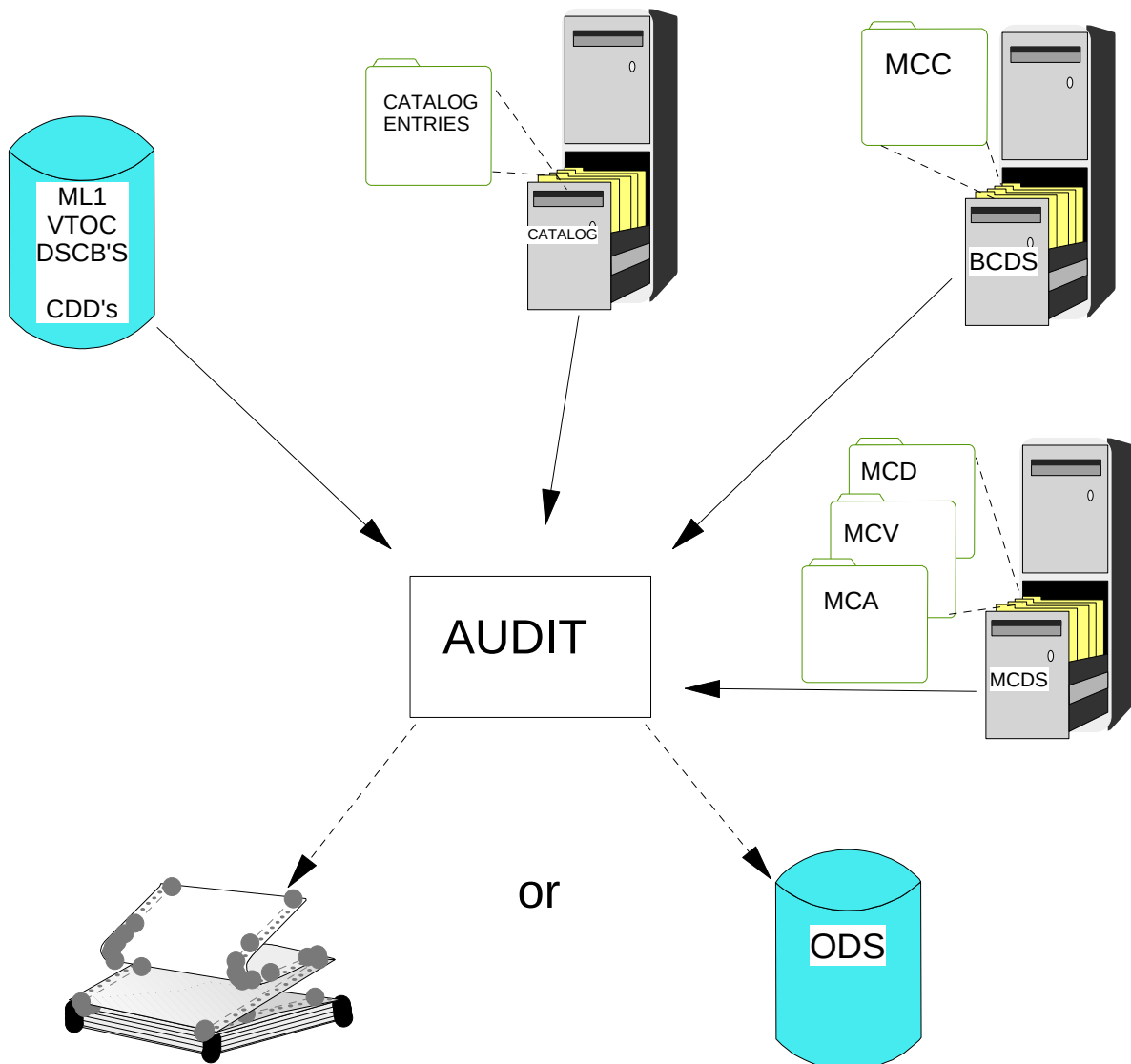
- TTOC structure is verified. For each data set in the TTOC, the following checks are made:
 - For ML2 volume:
 - MCD record
 - MCA record
 - Catalog entry indicates MIGRAT
 - For backup volume:
 - MCB record
 - MCC record

When to use this command:

- After recovery of the BCDS or MCDS
- Recall failure from ML2

Notes:

AUDIT MEDIACONTROLS VOLUMES(ml1ser)



```
AUDIT MEDIACONTROLS VOLUMES(ml1ser ...)
                        FIX | NOFIX
                        OUTDATASET(dsname) | SYSOUT(class)
                        REPORT(ERRORS)
                        SERIALIZATION(DYNAMIC | CONTINUOUS)
```

The MCV record is found for the volser specified in the command then the ML1 VTOC is read and the following is performed:

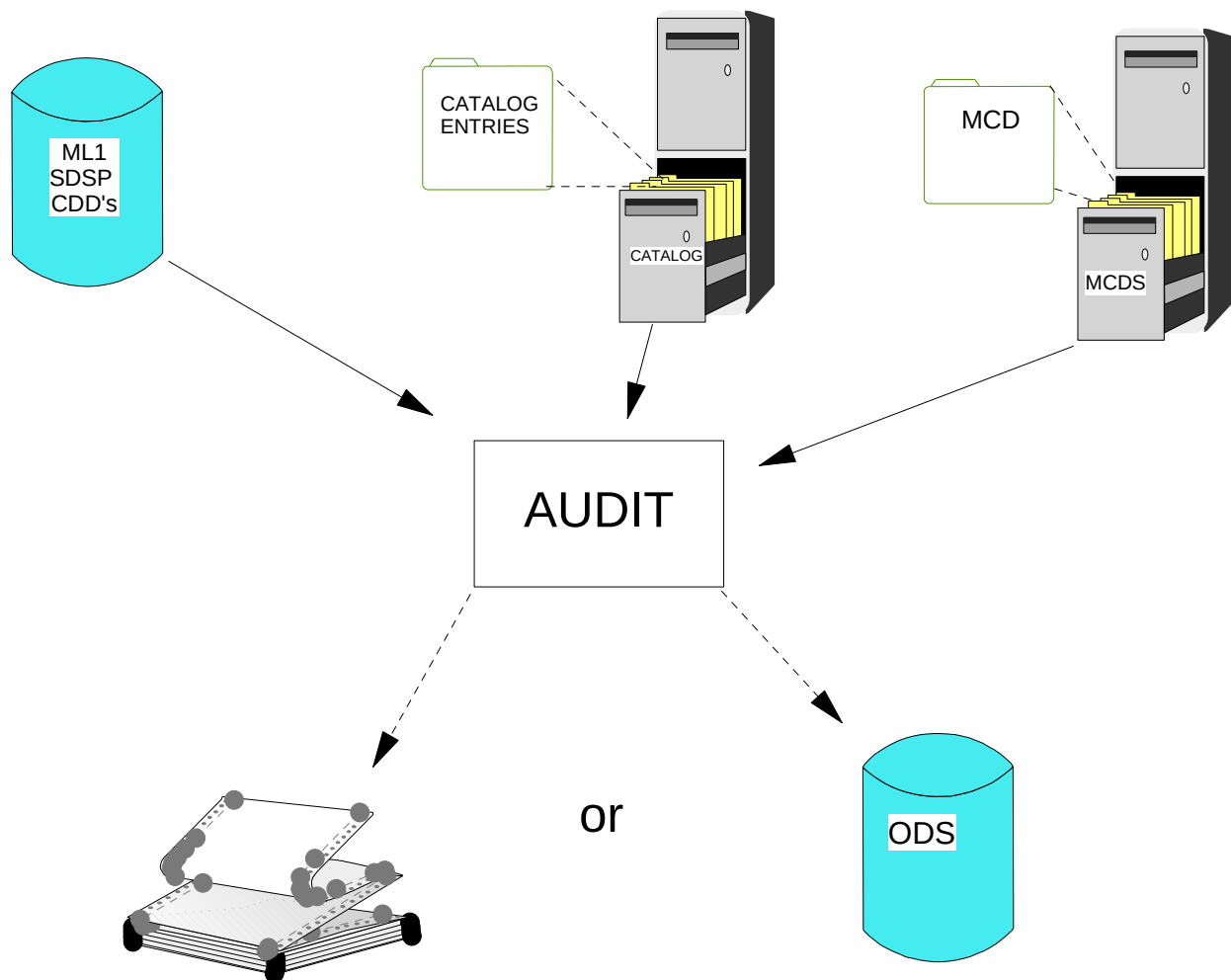
- If a backup version is found the following checks are performed:
 - MCC record
 - CDD record
- If a migrated version is found the following checks are performed:
 - MCA record
 - CDD record
 - MCD record
 - Catalog checked to make sure VOLSER=MIGRAT

When to use this command:

- When directed by the output of another AUDIT

Notes:

AUDIT MEDIACONTROLS(SDSP) VOLUMES(ml1vols)




```
AUDIT MEDIACONTROLS(SDSP) VOLUMES(ml1vols ... )  
      FIX | NOFIX  
      OUTDATASET(dsname) | SYSOUT(class)  
      REPORT(ERRORS)  
      SERIALIZATION(DYNAMIC | CONTINUOUS)
```

SDSP records are read sequentially, when a CDD record is found the following checks are made:

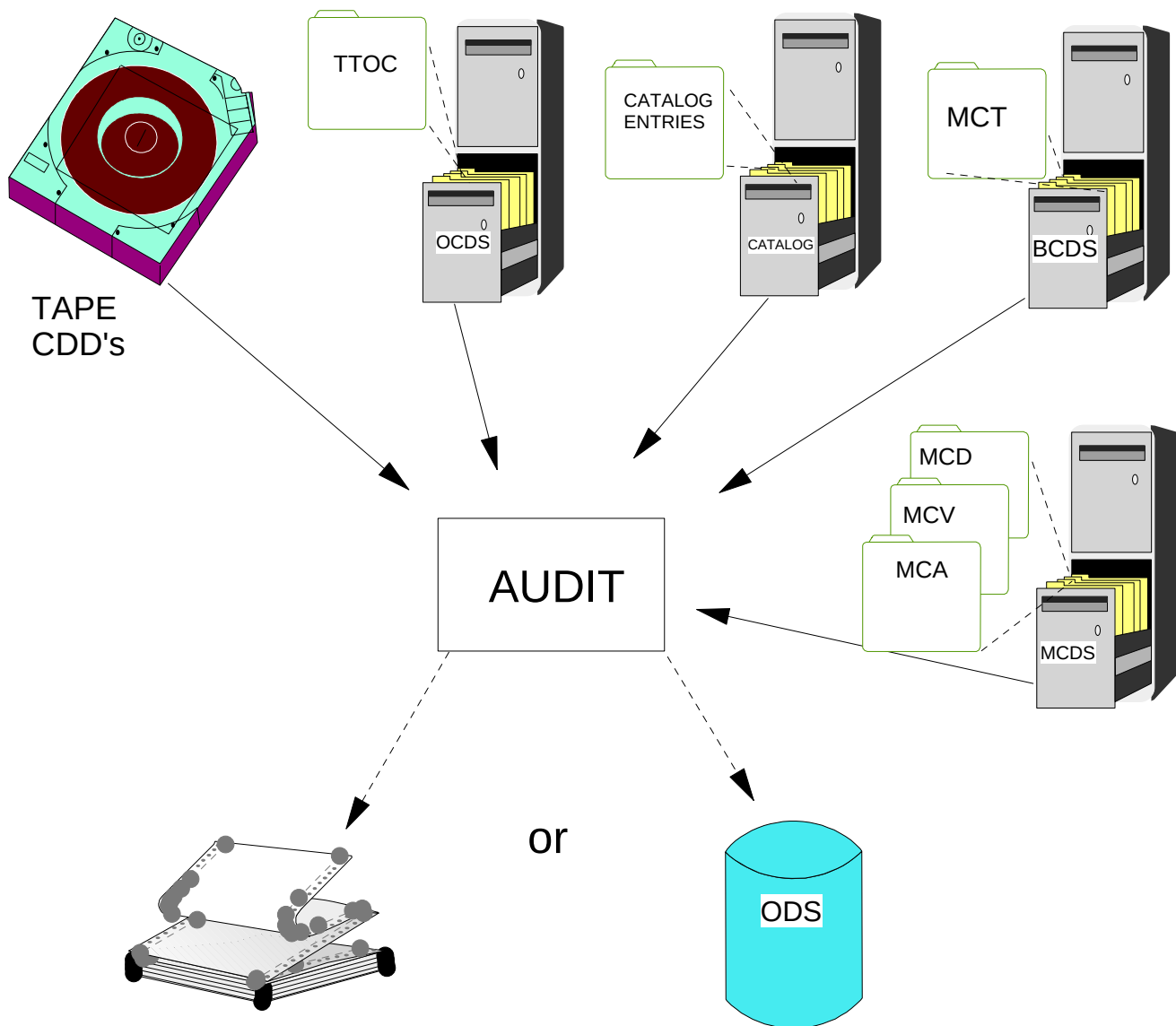
- MCD record
 - Note: SDSP migrated data sets do not have MCA records
- Verify that the data set's catalog entry indicates volume = MIGRAT

When to use this command:

- After recovery of MCDS
- After recovery of an ML1 volume

Notes:

AUDIT MEDIA CONTROLS VOLUMES(tapser)



```
AUDIT MEDIACONTROLS VOLUMES(tapser ... )  
      FIX | NOFIX  
      OUTDATASET(dsname) | SYSOUT(class)  
      REPORT(ERRORS)  
      SERIALIZATION(DYNAMIC | CONTINUOUS)
```

The specified TAPE(s) is read sequentially and the following is performed after verifying that an MCV or MCT record exists:

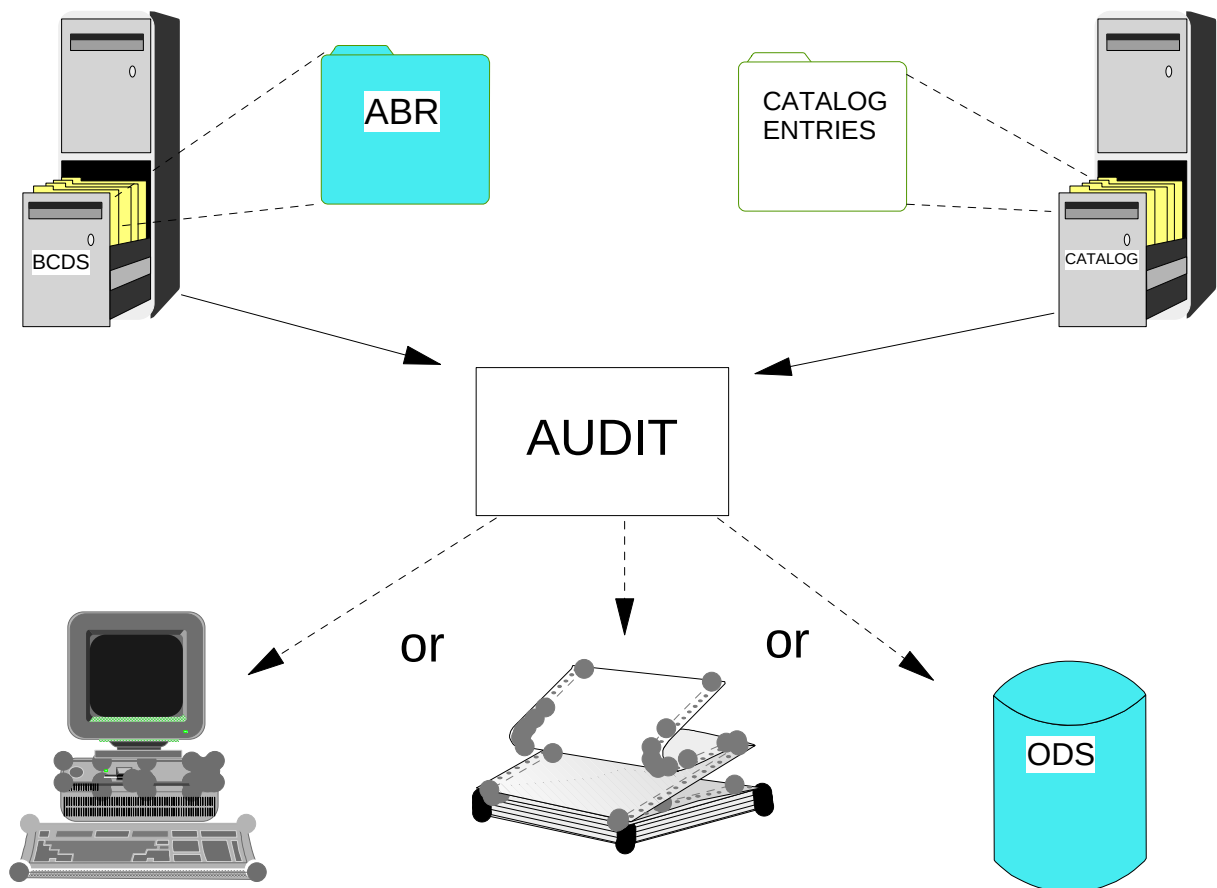
- The following checks are performed for every CDD that is found:
 - For an ML2 volume:
 - ★ Catalog checked to ensure that the data set indicates VOLUME=MIGRAT
 - ★ MCD record
 - ★ Block count for the data set (FBID).
 - For a backup volume:
 - ★ Block count for the data set.
- After the tape has been read the following checks are performed:
 - Valid block count
 - Total block count

When to use this command:

- When directed by the output of another AUDIT
- Repair a TTOC entry for an ML2 volume
- Repair a TTOC entry for a BACKUP volume
- Recovering an ML2 volume
- After receiving an ARC0378I message

Notes:

AUDIT ABARSCONTROLS (aname)



AUDIT ABARSCONTROLS (agname)
OUTDATASET(dsname) | SYSOUT(class) | TERMINAL
SERIALIZATION(DYNAMIC | CONTINUOUS)

The ABR records for the aggregate specified are read sequentially and the following checks are made:

- The following checks are performed for each ABR record that is found:
 - The control file is cataloged
 - The data file(s) generated by the aggregate are cataloged
 - The instruction file is cataloged (if one was created)

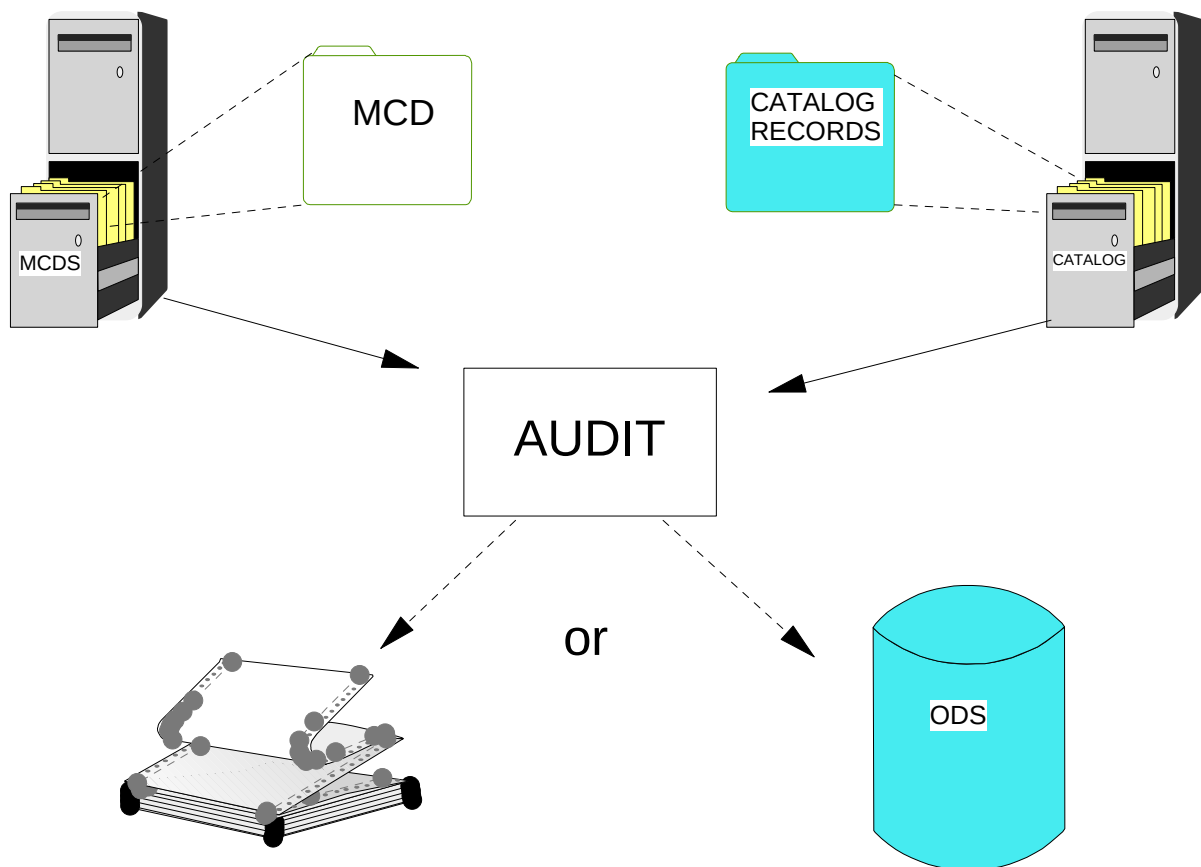
DFSMSHsm only

When to use this command:

- after a recovery of the BCDS
- after the recovery of a catalog
- periodically even when no errors are detected

Notes:

AUDIT MASTERCATALOG



```
AUDIT MASTERCATALOG |      NOFIX  
    USERCATALOG(catname) OUTDATASET(dsname) | SYSOUT(class)  
                        REPORT(ALL | ERRORS)  
                        SERIALIZATION(CONTINUOUS)
```

The Catalog is read and the following checks are made:

- If VOLSER = MIGRAT, ensure that the MCD record exists

When to use this command:

- After recovery of a user or master catalog

Notes:

Sample

AUDIT MEDIACONTROLS

Output

```
-- DFHSM AUDIT -          ENHANCED AUDIT -- LISTING - AT 18:55:40 ON 94/10/27 FOR SYSTEM=3090
COMMAND ENTERED:
AUDIT MEDIACONTROLS VOLUMES(A00021) FIX ODS(SAMPLE.MEDIA.CONTROLS)

/* ERR 51 A00021 - V RECORD INDICATES TAPE, NO TTOC EXISTS
/* FIXCDS T L2-A00021-0000 CREATE
/* FIXCDS T L2-A00021-0000 PATCH(X'00000002' BITS(...1...))
/* FIXCDS D JEFF.C2 VERIFY(X'00000000' A00021) PATCH(X'00000024' X'0094300F')
/* FIXCDS D JEFF.C2 VERIFY(X'00000000' A00021) PATCH(X'00000007' BITS(0.....))
/* FIXCDS T L2-A00021-0000 CREATE(X'00000034' X'40') ENTRY(DFHSM.HMIG.T463510.JEFF.C2.I4300)
/* FIXCDS T L2-A00021-0000 ENTRY(DFHSM.HMIG.T463510.JEFF.C2.I4300) PATCH(X'00000030' X'5E0110')
/* FIXCDS T L2-A00021-0000 ENTRY(DFHSM.HMIG.T463510.JEFF.C2.I4300) PATCH(X'00000035' X'000000')
/* FIXCDS T L2-A00021-0000 PATCH(X'0000001E' X'404040404040')
/* FIXCDS T L2-A00021-0000 ENTRY(DFHSM.HMIG.T463510.JEFF.C2.I4300) PATCH(X'0000002C' X'00000001')
/* FIXCDS T L2-A00021-0000 PATCH(X'0000000E' X'0001')

/* FIXCDS T L2-A00021-0000 VERIFY(X'00000010' X'00000000') PATCH(X'00000010' X'00000001')
/* FIXCDS T L2-A00021-0000 VERIFY(X'00000014' X'00000000') PATCH(X'00000014' X'00000001')
/* FIXCDS T L2-A00021-0000 VERIFY(X'0000000C' X'0000') PATCH(X'0000000C' X'0001')
- END OF -          ENHANCED AUDIT - LISTING -
```


The body of the report normally consists of two lines printed for each error detected. The top line gives you the error code and a brief description of the error. The line immediately following the error code will list the appropriate repair command, whenever AUDIT is able to identify it. Be aware that not all actions performed when the FIX parameter is specified appear in an executable form in the AUDIT output data set. For this reason, you cannot expect to get the same results by running the AUDIT first with the NOFIX option to an ODS, then editing the output data set and submitting it as a CLIST.

Notes:

I still have more
questions.... Like..
What do the ERR s mean?



Figure 26. Error Codes Used in AUDIT Reports		
Description	Audit Repair Action	Troubleshooting Hints

*ERR 51 volser - V RECORD INDICATES TAPE, NO TTOC EXISTS		
A migration (v) record for volume volser exists, but no TTOC base record is found.	- If the volume record indicates - A tape other than one in single-file format, and - An empty volume - an ADDVOL command is generated for volser. - If the volume record indicates a tape in single-file format, the base TTOC is created, and individual data set TTOC entries are created as the tape is read.	

For more information read:

Appendix A in

DFHSM System Programmer's Command Reference

or

DFSMS/MVS 1.1 Storage Administration Reference for DFSMShsm

Notes:

Diagnostic Value of the AUDIT Command

The AUDIT command has the ability to detect, report, and provide repairs for discrepancies between VTOCs, catalogs, and the CDSs. Since HSM relies so heavily on the VTOCs, Catalog, and CDSs, it is essential that they remain error free. Thus, the AUDIT command gives an authorized user the ability to detect errors before they can cause damage